8.3 Analyzing Geometric Sequences and Series

Name _____ Period ____ Date _____

Algebra II 8.3 · Student sheet

I can…

- tell whether a sequence is geometric and write its rule.
- write a rule from one term and the common ratio, or from two terms.
- find the sum of a finite geometric series.

Vocabulary · fill in the missing word

Geometric sequence — the ratio of _____ terms is constant

Common ratio r — the _____ ratio

Rule — a_n = _____

Sum of a finite geometric series — S_n = _____, $r \neq 1$

Part 1 · Spreading the news

On day 1 you tell 3 friends a secret. Each day, everyone who heard it the day before tells 3 new people.

Day	1	2	3	4	5	6
New people						
Total who know						

Rule for the new people on day n: _____

Part 2 · Notes: fill in the blanks

Geometric: the _____ of consecutive terms is constant (r).

Rule: a_n = _____

Sum: S_n = _____

Part 3 · Examples (work them with the class)

Example 1

Tell whether each sequence is geometric.

a. 5, 10, 20, 40, … b. 1, 4, 9, 16, … c. 96, −48, 24, −12, …

Example 2

Write a rule for the nth term of 4, 20, 100, 500, … Then find a_7.

Example 3

Two terms of a geometric sequence are $a_2 = 12$ and $a_5 = -768$. Write a rule for the nth term.

Example 4

Find $\sum_{i=1}^{8} 4(3)^{i-1}$.

Example 5

A scholarship fund gives $2000 the first year and 5% more each year after that. How much does it give in all over 10 years?

Part 4 · Partner practice

Partner 1

One term of a geometric sequence is $a_4 = 12$, and $r = 2$. Write a rule for the nth term.

Partner 2

Find the sum of the first 6 terms of $1, -2, 4, -8, \ldots$